Statistics for Computer Science

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Why do we want to study statistics?

- Underpins many of the techniques we will use in machine learning and artificial intelligence.
- The world into which we deploy our AI-enabled systems is uncertain and ever changing.
- Statistics is the language of data science. It provides the tools to understand this world and hence design better systems and understand the ways in which they fail.

We will look at:

- *Descriptive statistics* and *data visualization*
- *Probability*.
- *Data correlation* and *regression*

Descriptive Statistics and Data Visualization

A survey was carried out to find out the ages of people visiting a news website. One hundred people were surveyed and the ages recorded are presented in a table below with results placed into ascending order for convenience.

19	19	19	19	20	20	21	21	21	23
24	24	25	25	26	26	28	28	29	31
31	32	32	33	34	35	35	36	37	38
38	38	39	39	40	41	41	41	42	42
43	43	44	45	45	46	46	47	50	52
53	53	54	55	56	56	56	57	57	59
59	60	61	61	61	61	64	64	64	66
67	67	68	68	69	69	70	70	70	71
71	72	74	75	75	75	76	77	77	77
77	78	78	78	79	79	79	79	80	80

A first step in understanding the table may be to *group data*. The question we want to ask of the data informs how that grouping might happen.

"Is our website equally popular across all age groups.".

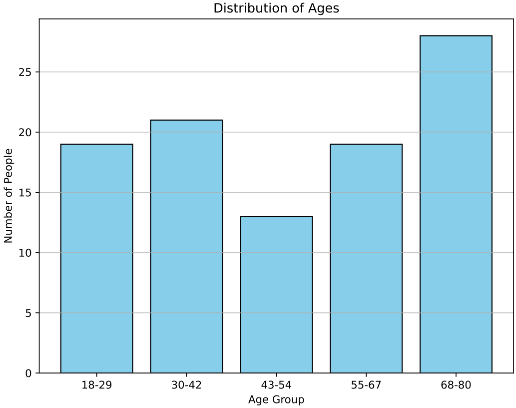
In this case then we might choose a number of age groups as:

$$(18-29),(30-42),(43-54),(55-67),(68-80)$$

and then summarise our data into a smaller set of values:

Age Group	Frequency
18-29	19
30-42	21
43-54	13
55-67	19
68-80	28

We might then make use of a visualization to show these results. In this case a histogram seems like a sensible choice.



Looking at the data in this way shows us that we attract many people in the oldest group but we are not successful in reaching people in the 43-54 age range.

Central Tenancy

A common requirement in handling data is to find out what the average "thing" is. I'm being deliberately vague here, but it might be the average age of a patient, the aver temperature at which failure occurs for a jet engine etc.

Consider a machine which makes ball bearings. We know they are all very similar but no two ball bearings are identical. If we consider their weight we could describe all the ball bearings as have

$$\text{weight} = \bar{w} \ \pm \ \epsilon$$

where $\bar{w}$ is the weight we designed the machine to create and ϵ is some measure of variance on that value.

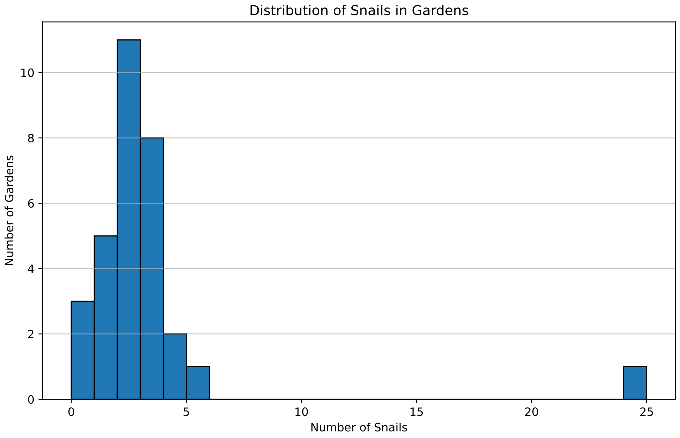
Of course if we were involved in the design of the machine we might know precisely what $\bar{w}$ was but if the process is natural finding this value hidden by noise is more tricky.

The most common measures one would use to find this value are the **mean**, **mode** and **median**. Each of these are considered to be measure of central tendency.

Let's consider the following set of data which reports the number of snails found in 31 gardens one a one hour time period. Rather than providing the raw data as before we now present the data as a frequency table.

Number of Snails	0	1	2	3	4	5	24
Frequency	3	5	11	8	2	1	1

This plot shows a distinct peak meaning it's most common to see 2 snails and the likelihood of seeing more or less snails reduces as we move away from this point. We also see an *outlier*. An outlier is a sample (or data point) which doesn't look like the rest (24 snails in one garden).



Question 1.

What might be the cause of the outlier?

So let's start with our question:

"How many snails, on average, will you find in a garden"

Mode, Median and Mean all tell us something about the "centre" of the data.

Mode is simply the "most common" number in the data. For our data we simply look for the number with the highest frequency which is 2, i.e. 11 gardens reported 2 snails. This means that if we put all 31 numbers into a bag and picked one at random the number we are most likely to see is 2.

Median is the middle number when all the numbers are placed in numeric order. For our data we would then write the data out as follows:

$$0, 0, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2 \cdots$$

That is we will write 3 0's, 5 1's, 11 2's etc. In total there will be 31 gardens (data points), so the median will return the 16th number. In this case the 16th number is a 2 so the median = mode = 2.

Note: If we had only 30 gardens we would select the middle 2 gardens (15 and 16) add their values together and divide by 2.

Finally let's look at the **mean**. This is probably the most common measure of centrality and is calculated by adding all the results recorded together (the number of snails in each garden) and dividing by the number of instance (gardens). This can be written mathematically as

$$\mathrm{mean}(\mathbf{x}) = \left(\sum_{i=1}^{n} x_i \right) / n \tag{1}$$

where n is the number of gardens and x_i is the number of snails in garden i.

When a frequency table is provided we can use the following equation:

$$\text{mean}(\mathbf{x}) = \left(\sum_{i=1}^{m} (x_i \times f_i) \right) / n \tag{2}$$

where m is the number of groups and f_i is the frequency within each group.

For our problem then the mean is

$$((3*0)+(5*1)+(11*2)+(8*3)+(2*4)+(1*5)+(1*24))/31 = 2.84$$

mode = **median** = 2

mean = 2.84

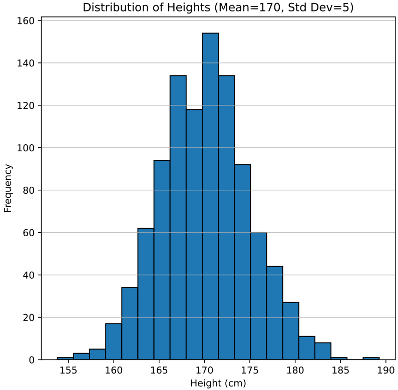
We can see that the mean that is reported is higher than the mode and the median.

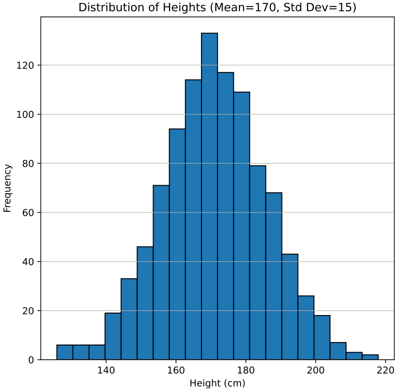
Question 2.

1. Which of the values is a better measure of centrality?
2. What is the value of the mean if we exclude the outlier?
3. Should we exclude the outlier?

Dispersion

Two researchers were asked to gather a sample of participants heights. Both report a **mean** of 170cm.

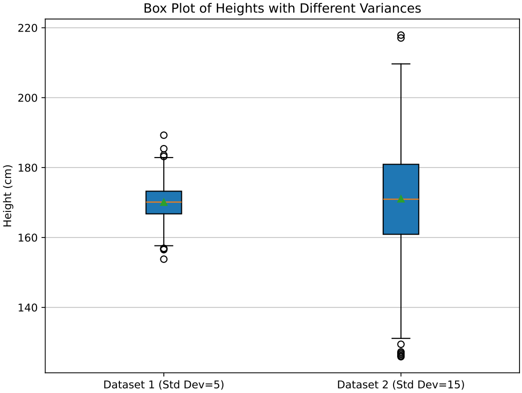




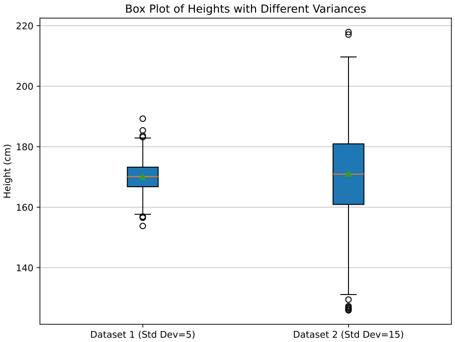
Box plots

Another way of viewing this data is using a box plot (or more accurately a box and whisker plot).

This shows clearly that data set one has narrower range of values than data set two.



Interquartile range (IQR)



The range covered by the box is the middle 50% of your data the two whiskers then tell us about the other data, the lower quarter and upper quarter.

Circles are commonly plotted to represent *outliers*.

Standard Deviation

A second method for describing the *dispersion* of a dataset is called the standard deviation and it is calculated as follows:

$$s = \sqrt{(\frac{\sum_{i=1}^{n}(x_i^2)}{n} - \bar{x}^2)} \tag{3}$$

Note x_i is each of the height we have recorded in our experiment.

- We square each height in the list and then add them together.
- Next we divide by n, the number of heights we recorded.
- Finally we take away the mean $\bar{x}$ squared.

So this equation then says the standard deviation (s) is the square root of the mean of squares ($\frac{\sum_{i=0}^{n}(x_i^2)}{n}$) minus the square of the mean ($\bar{x}^2$).

Let's give it a go on a small set of data

x	x^2
2	4
3	9
5	25
6	36
8	64
$\bar{x} = 4.8$	$\sum(x^2) = 138$

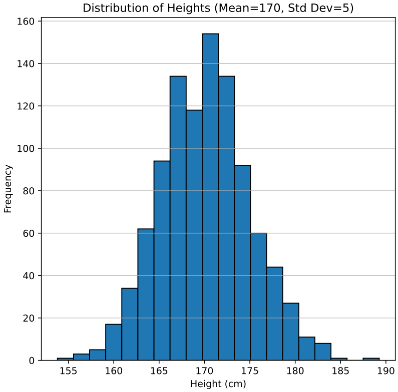
so

$$s^2 = \frac{138}{5} - (4.8)^2 = 4.56$$

$$s = \sqrt{(4.56)} = 2.14$$

So what does this number actually tell us?

Well without going into too much detail it says the majority of the data will lie within 2 standard deviations of the mean. So looking once again at our histograms we see that most of the data should lie within the range 170 ± 10 for the left hand graph and within 170 ± 30 for the right hand graph.



Question 3.

We sampled the output from two machines which create biscuits. For each biscuit we measured it's weight. Calculate the mean and standard deviation of biscuit weights for each machine and then comment on the performance of each machine[a].

Machine A									
196	198	198	199	200	200	201	201	202	205
Machine B									
192	194	195	198	200	201	203	204	206	207

[a] Question taken from A concise course in Advanced Level Statistics with worked examples. J. Crawshaw and J. Chambers.

Probabilities

The nature of a chance event is that we can not say what will happen, but for some systems, and with appropriate assumptions, we can predict accurately the *proportion* of times an event would occur.

> *Consider a normal six sided dice. If I throw the dice 600 times how many sixes am I likely to get?*

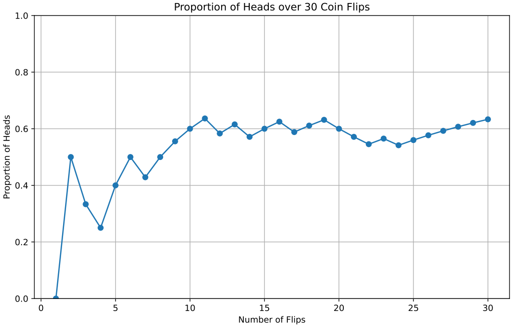
For our dice for example we can reasonably say that all of the numbers are equally likely (assuming a fair dice) and that therefore we will get a 6 (or any other number) $\frac{1}{6}$ of the time we roll our dice.

Question 4.

Take a coin and flip it 10 times. Fill in the table below with pHeads and pTails being the proportion of heads and tail you have flipped so far.

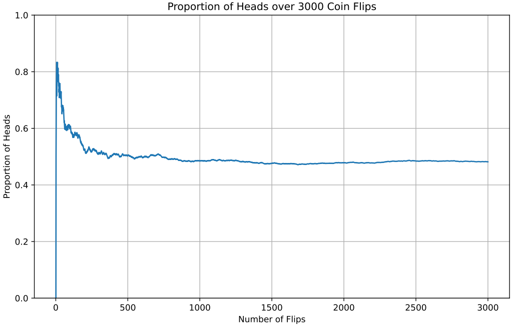
Flip No.	Outcome	pHeads	pTails
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

I ran the same experiment but repeating the experiment 30 times. The first few flips were as follows: $\{T, H, T, T, H, H, \cdots\}$



Notice that at the beginning the proportion varies greatly but as the experiment continues the variation decreases. Notice also that we also still seem to be quite a way off of 0.5!

If I run the experiment again, Figure **??** this time for 3000 flips we can see the value setting to the number we expect. However we see in this experiment that the initial estimate is very high.



When we write a probability then we are saying the proportion of times we would expect and event to occur if the event were repeated infinitely often.

For our example the event E is:

$$E = \text{coin lands on heads}$$

and the probability of the event is:

$$P(E) = 0.5$$

Question 5.

1. Can you think of a way of describing the following probabilities using plain English statements.

- $P(E) = 0$
- $P(E) = 1$

2. Can a probability be negative?

A quick note. **Events** and **Outcomes** are not the same thing. Consider a dice roll the outcomes are six different numbers we can get.

The event however is the subset of outcomes we are interested in. So each of the following could be an event:

1. Rolling a six : $P(E) = \frac{1}{6}$
2. Rolling an even number : $P(E) = \frac{1}{2}$
3. Rolling a number greater than 4 : $P(E) = \frac{1}{3}$

Aside: Although we have started this discussion of probabilities using dice and coins it is worth noting that probabilities are used in situations where calculating probabilities in advance of the experiment is not quite so easy.

Consider the following examples.

1. **Weather:** Will it snow this Christmas?
2. **World Events:** Will the U.S.A. have another Civil War?

Sometimes probabilities can be based on the frequency of easily observed events (or easily modelled processes) but sometimes probabilities are a way of communicating our own beliefs. For both cases probabilities are often stated without making it clear what assumptions are made, this can be problematic. For example is the frequency of previous snow a good model for future snow?

Multiple events

Often we are interested in not just one event but multiple events happening either at the same time or in sequence.

For a moment think about how would you might approach calculating the following probabilities.

1. Getting two heads when flipping two coins
2. Rolling three sixes on three dice
3. Getting at least one six when 3 dice are rolled
4. If we select 3 cards at random from a standard deck of cards at least one of them will be a picture card?

Two Coins

One way to approach this problem is to list all the possible outcomes and then work out the proportion which match our desired event.

- Head, Head
- Head, Tail
- Tail, Head
- Tail, Tail

Notice that it is importance to be systematic in writing out the possible outcomes. Here it is clear that the probability of two heads is $P(E) = 0.25$ but if I had written $\{$ 2 Heads, 2 Tails, 1 Head and 1 Tail $\}$ as my list of outcomes I may have missed that there are four different combinations possible.

3 Dice

For our 3 dice problem there are 216 ($6 \times 6 \times 6$) different outcomes, I really don't want to write all of these out.

Fortunately I can use a **multiplication** rule for this problem because I know that rolling one dice does not affect the next roll, i.e. the events are *independent*.

This means that we can calculate the total probability of each event (getting a six) and then multiplying these together so that

$$P(3 \text{ sixes}) = \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216}$$

So generally if we can be sure that two events (E and F) are independent we can simply multiply the event probabilities i.e. $P(E \text{ and } F) = P(E) \times P(F)$.

Question 6.

Calculate the probability of obtaining a total of 3 in each of two successive rolls of 2 dice.

Step 1 : Probability of rolling 3 on two dice

Step 2 : Calculate the probability of doing this twice in a row.

At least 1 six

Probabilities always add up to one so we can use this to answer this question by saying we want 1 minus the the probability of getting at NO sixes.

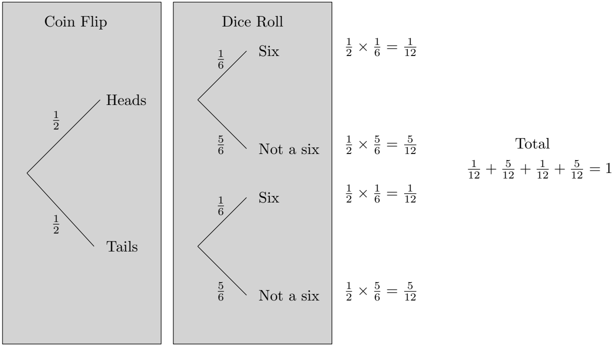
$P(\text{no sixes}) = \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = 0.579$

Now we know that all other options will have at least one 6 so:

$P(\text{at least one six}) = 1 - P(\text{no sixes}) = 1 - 0.579 = 0.421$

ALL possible events MUST add to 1.

Let's introduce a graphical device called a tree diagram.



For each experiment we walk along a branch of the tree multiplying the probabilities are we go. If we then add of the final outcomes we get a sum of 1.

Picture Cards

So far every experiment has made use of the fact that the outcomes are independent, but what if they are not. Well let's start by thinking about the Picture Card problem and what might happen.

> *If we select 3 cards at random from a standard deck of cards at least one of them will be a picture card?*

Why isn't this problem just like the others?

Once we take a card out of the deck then the deck has changed. The first action we took impacts the probability of the second. So that means that

> *The first card we draw will be a picture card with probability* $\frac{3}{13}$

But this isn't our only chance to draw a picture card so now we need to work out the following ...

Question 7.

What is the probability of drawing a picture card with the second card?

Regardless of what we drew the first time we only have 51 cards in the deck. But also if we picked a picture card the first time then the number of picture cards will go down.

This is another example of where we might use our trick of looking for the opposite of what we want as it reduces the number of options. So Let's change our question a little :

> *If we select 3 cards at random from a standard deck of cards NONE of them will be a picture card?*

So now the probability of not drawing a picture card each time is as follows:

$$P(\text{no pic}) = \frac{40}{52} \times \frac{39}{51} \times \frac{38}{50} = 0.447$$

Now we can answer our original question by assuming all other events will include a picture card.

$$P(\text{pic}) = 1 - P(\text{no pic}) = 1 - 0.447 = 0.553$$

Bayes Theorem

This last example introduces us to the idea of asking a probabilistic question when we know the outcome of a previous probabilistic event.

Why is this important? Well consider a medical scenario in which we can observe symptoms and we want to assess the probability of a disease. In Natural language we might want to write something like:

> *The probability of someone having a cold given that they have a runny nose is 0.4 whilst the probability of them have cold if they don't have a runny nose is 0.2.*

Mathematically you would write the statement "The probability of A given the event B happened" as:

$$P(A|B)$$

This is known as a conditional probability since the probability of A is a condition of B. For our case we might write:

$$P(\text{cold}|\text{runny nose}) = 0.4$$

and

$$P(\text{cold}|\neg\text{runny nose}) = 0.2$$

notice that these don't add up to one because that's not all the possible outcomes and combinations.

One question of course is where did those numbers come from? In this case it might be experimental (or observed data) but more generally it may not be possible to have numbers directly for these conditional probabilities.

Consider the following statements:

- *Probability of a particular form of cancer being present is 0.01*
- *Probability of a positive test when cancer present is 0.9*.
- *Probability of a positive test when cancer not present is 0.05*.

But our question is "if you test have a positive cancer test how likely is it that you have cancer?". We don't have a value for this.

Writing this formally we want to know:

$$P(\text{cancer}|\text{positive})$$

Fortunately there is a way of calculating this value and it uses something called "Bayes Rule":

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} \tag{4}$$

Let's now plug in our values and see what we get

$$P(\text{cancer}|\text{positive}) = \frac{P(\text{cancer})P(\text{positive}|\text{cancer})}{P(\text{positive})}$$

Looking at our data in above we see that:

- $P(\text{cancer}) = 0.01$
- $P(\text{positive}|\text{cancer}) = 0.9$

We still need to know $P(\text{positive})$, the probability of receiving a positive text result. Let's consider 10,000 patients selected at random.

Question 8.

Given the probability data above, answer the following questions:

- a = Out of 10,000 people how many will have cancer
- b = How many with cancer will get a positive test
- c = How many without cancer will get a positive test

What is the value of $P(\text{positive}) = \frac{(b+c)}{10{,}000}$.

Using this calculated value, what is

$$P(\text{cancer}|\text{positive}) = \frac{P(\text{cancer})P(\text{positive}|\text{cancer})}{P(\text{positive})}$$

How would you interpret this value?

Data correlation and regression

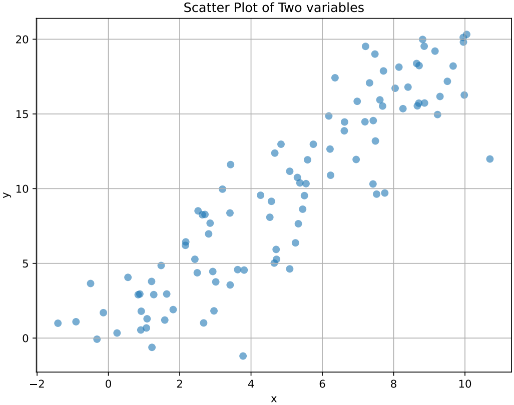
One of the problems we are often looking at in AI is how variables are related. We want to know if a given factor has an affect on an outcome, or to put it in simpler terms is a thing we can control will change something we are interested in and if so how.

So we might ask questions like does the speed we drive at impact the severity of injuries in car accidents, or does the amount of exercise we undertake impact bone health in later life.

Questions like this require us to consider if variables are **correlated**.

Let's start by looking at some data.

You might imagine here that x is age and y is salary for example or amount of exercise and additional years of healthy life.



Visually we can see that as x increases so does y. This is known as a positive correlation.

Question 9.

It has been shown that butter consumption is positively correlated with ticket prices at North American movie theatres[a].

Can you explain this correlation?

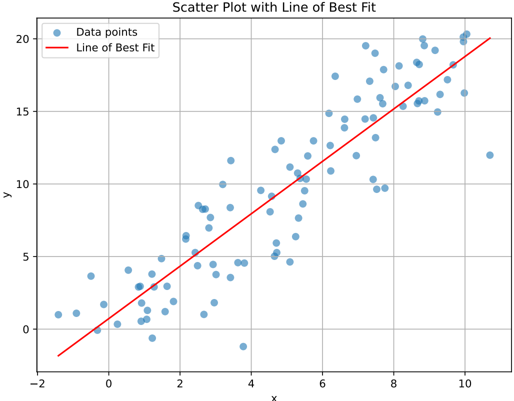
[a] https://www.tylervigen.com/spurious/correlation/1604_butter-consumption_correlates-with_ticket-prices-at-north-american-movie-theaters

We are often interested in the function which describes one variable in terms of the other.

If we assume the data is like a straight line we can described it as:

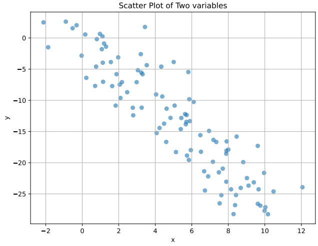
$$y = mx + c$$

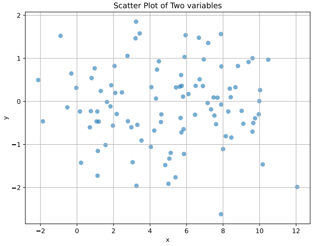
where m is the gradient of the line and c is the point where the line crosses the y axis.



Question 10.

Using a pen and ruler draw a line of best fit on the two scatter plots.





Least Squares Regression

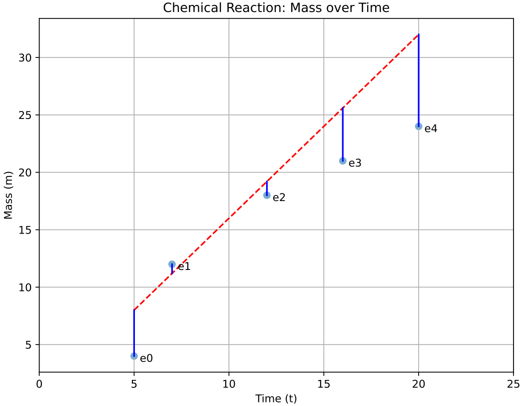
To round off this course we are going to look at a method of calculating the best fit line using something called least squares regression.

Let's start by looking at a small set of data:

Time, x (min)	5	7	12	16	20
Mass, y (g)	4	12	18	21	24

Table: Weight of a chemical over reaction time

We will start with a scatter plot of the data as shown in Figure **??**. Here we have drawn a line $y = 1.6x$. We don't know if this is the "best fit" but it's not a bad guess.



What we can now do is measure the distance of each point (vertically) from the line we have guessed (e_i). We think of these measurements as the *error* between our guess (line) and the observed data. The total error is then:

$$\epsilon = \sum_{i}^{n} e_i$$

where n is the number of data points (n).

It seems quite logical therefore to say that a good guess for the line would have a very small value of ϵ whilst a poor guess would have a large value.

Our job then is to find the line for which ϵ is as small as possible.

One process to find a function of best fit is to first think of a function that is a reasonable guess. For us this is a straight line, and the formula for this is commonly written as :

$$y = mx + c$$

So now we have to work out the values of m and c that give us the smallest error possible. One way we could do this is guess values and continue guessing randomly for 100 tries and choose the answer with the smallest value, but this seems a little inefficient.

In your AI course you will come across ways to do this for complex functions but for now we will look at a method using maths.

We know that gradient of a line b is the change in y divided by the change in x. What is maybe not so clear is that we can also calculate the gradient as

$$m = \frac{s_{xy}}{s_{xx}}$$

where s_{xx} is the variance of the values of x and s_{xy} is something called covariance.

Variance is a like the standard deviation we calculated earlier.

$$s_{xx} = \frac{\sum_{i=1}^{n} x_i^2}{n} - \bar{x}^2 \tag{5}$$

Note then that $s_{xx} = s^s$ and is similarly a measure of the "spread" of the variable x.

Covariance is still a measure of *joint* variability[1] and is calculated as follows:

$$s_{xy} = \frac{\sum_{i=1}^{n} x_i y_i}{n} - \bar{x}\bar{y} \tag{6}$$

So for our problem we have 5 data points let's make a small table to help up with our calculation of b.

[1] it's a lot like correlation

Question 11.

Complete the table and put the sum of each column in the final row.

x	y	x^2	y^2	xy
5	4	25	16	20
7	12			
12	18			
16	21			
20	24			
$\sum =$ 60	$\sum =$ 79			

Now we can calculate s_{xx}. We know that the average of x is $\bar{x} = \frac{60}{5}$ so,

$$s_{xx} = \frac{874}{5} - \left(\frac{60}{5}\right)^2 = 30.8$$

Similarly:

$$s_{xy} = \frac{1136}{5} - \frac{60}{5}\frac{79}{5} = 37.6$$

so the gradient of the correct line is

$$b = \frac{37.6}{30.8} = 1.22$$

Great we have a gradient, how do we now find c?

Well the trick is to know that the point $(\bar{x}, \bar{y})$ lines on the line. So we can rearrange our equation for a straight line to say:

$$c = \bar{y} - m\bar{x} = \frac{79}{5} - 1.22\frac{60}{5} = 1.16 \tag{7}$$

Question 12.

This felt like a lot of work so why might we want to work out the equation of a function that fits our data?

Of course just like other problems we have seen in this course we can get the computer to help us $\cdots$

1. Go to `www.wolframalpha.com`
2. Click on the examples button
3. Find the examples for Statistics
4. Find the Regression Analysis Examples
5. Click on the example starting “linear fit”
6. Click on the “$=$” button and explore the results
7. Try entering the data from our problem.
8. Do you get the same answer?